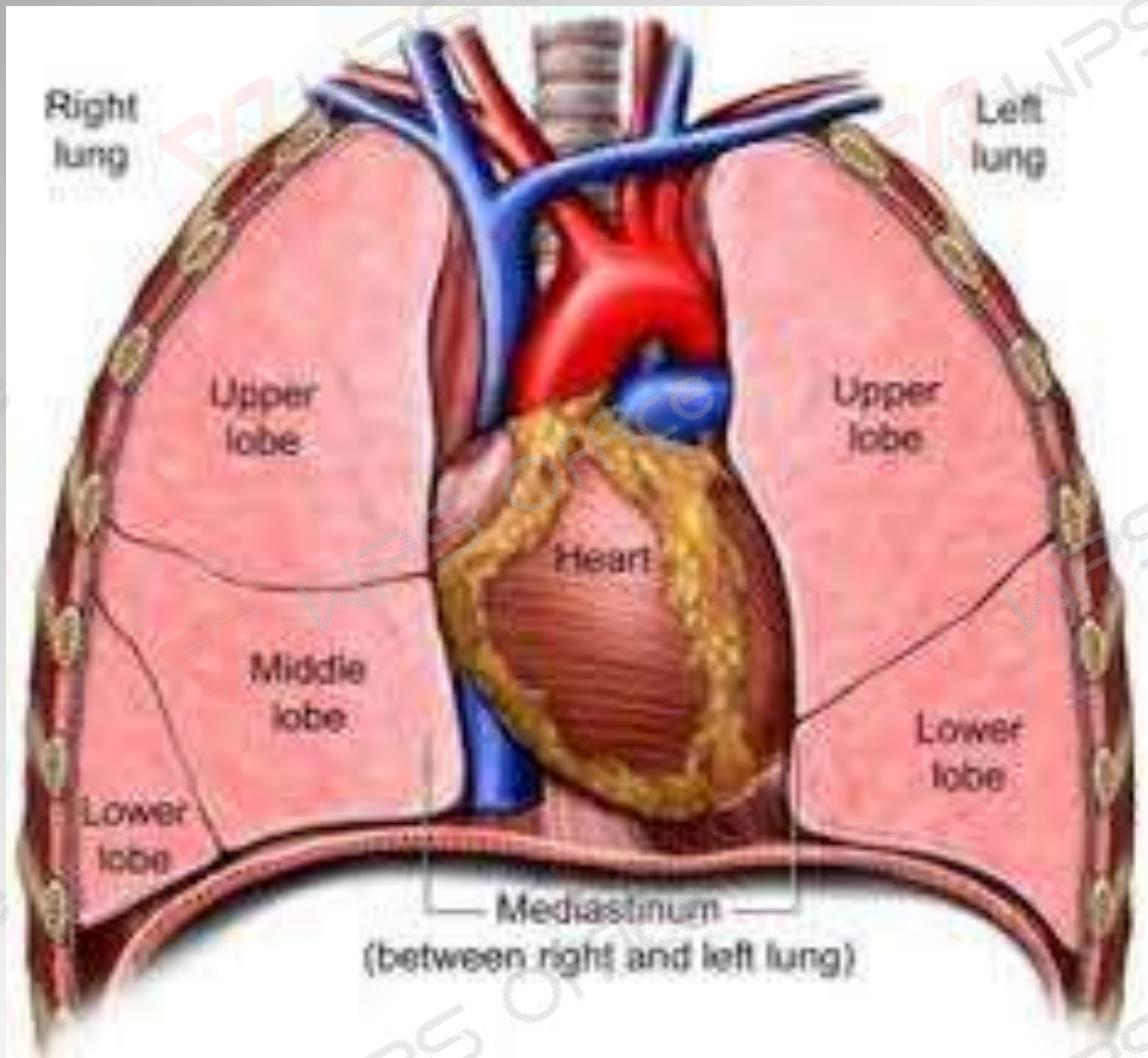
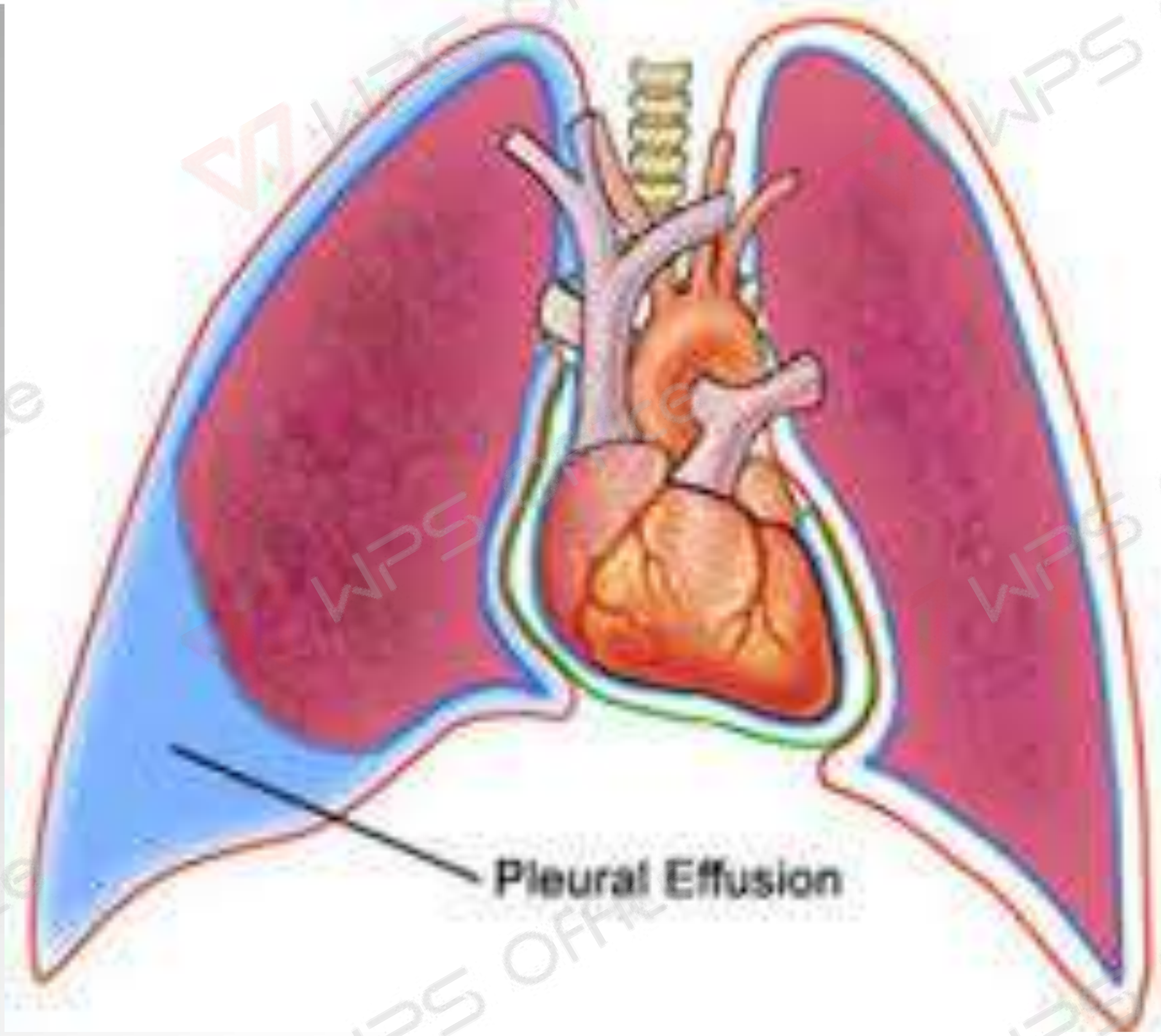
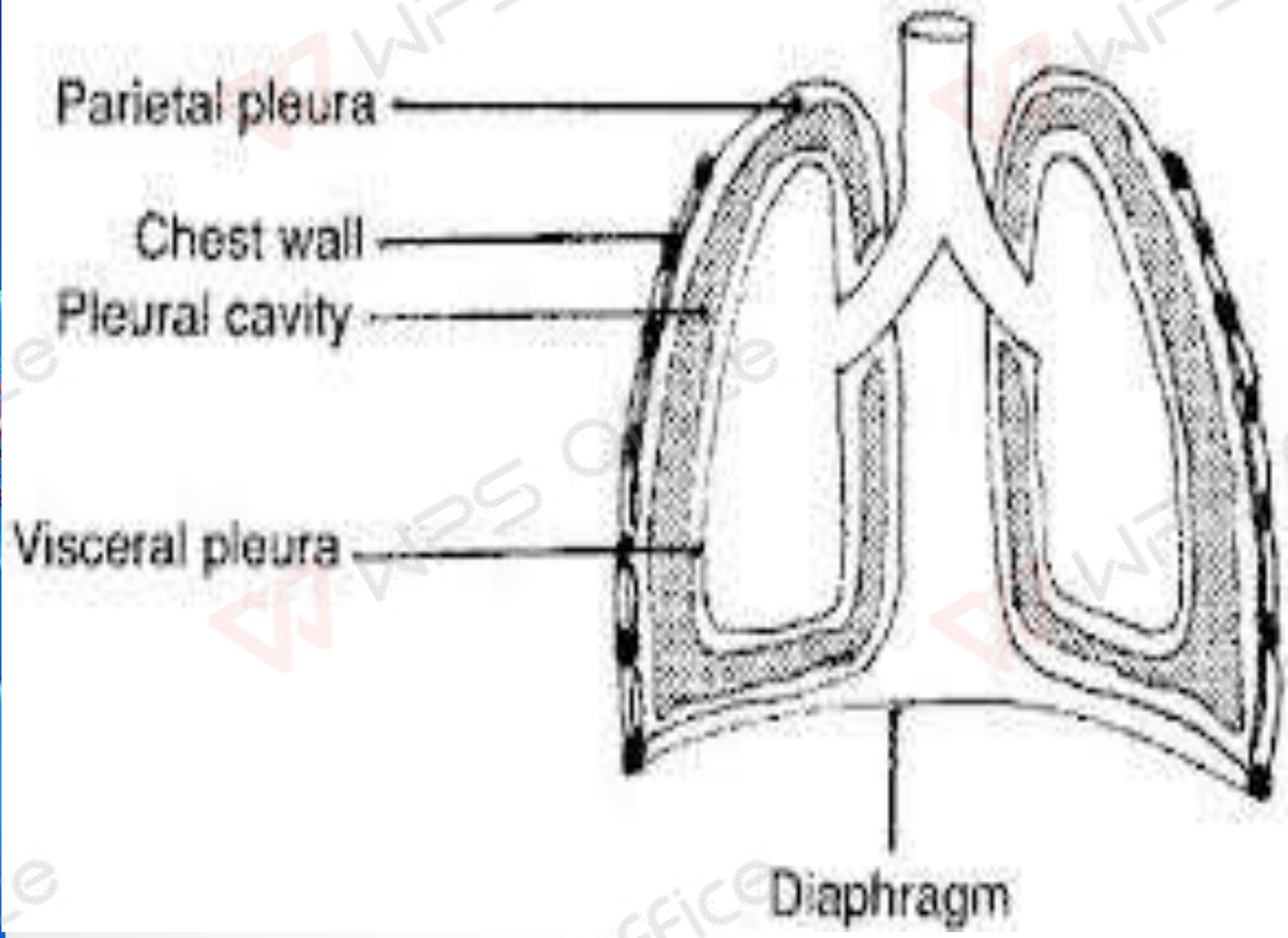
PLEURAL EFFUSION

Pr


Right lung
Left lung
Upper lobe
Upper lobe
Heart
Middle lobe
Lower lobe
Lower lobe
Mediastinum
(between right and left lung)


Pleural Effusion


Parietal pleura
Chest wall
Pleural cavity
Visceral pleura
Diaphragm

INTRODUCTION

The body produces pleural fluid in small amounts to lubricate the surfaces of the pleura, it lines the chest cavity and surrounds the lungs. The pleural cavity contains a relatively small amount of fluid, approximately 10 ml on each side .

Excessive Accumulation of fluid in the pleura

- *PLEURAL EFFUSION* is an abnormal, excessive collection of this fluid . Excessive amounts of such fluid can impair breathing by limiting the expansion of the lungs during respiration

According to Javed ansari

- **Pleural effusion**, sometimes referred to as “water on the lungs,” is the build-up of excess **fluid** between the layers of the **pleura** outside the lungs.
  The **pleura** are thin membranes that line the lungs and the inside of the chest cavity and act to lubricate and facilitate breathing.

Brunner and Suddarth’s

Con……………………………….

- **Pleural effusion is define as Build up of excess fluid between the layers of the pleura outside the lungs.**

Lippincott

- pleural effusion refers to a collection of fluid in the pleural space
- **luckmaan**

TYPES

- TRANSUDATIVE PLEURAL EFFUSIONS
- EXUDATIVE EFFUSIONS

Types of Effusions

TRANSUDATIVE PLEURAL EFFUSIONS

It caused by fluid leaking into the pleural space. This is caused by increased pressure in, or low protein content in, the blood vessels . A transudate is a clear fluid, similar to blood serum . It reflect a systemic disturbance of body

Causes of Transudates

- Atelectasis
- Cirrhosis
- Congestive heart failure
- Hypoalbuminemia
- Nephrotic syndrome
- Peritoneal dialysis

• ATELECTASIS

- is the collapse or closure of a lung resulting in reduced or absent gas exchange.

• CIRROSIS

- Hepatic Hydrothorax (**Pleural Effusion**) **Pleural effusions** complicate end-stage **liver disease** in 5% of patients. **Effusions** (defined as 500 mL or more of fluid within the **pleural** space) are typically right-sided. No cardiopulmonary **cause** for the **pleural effusion** is found.

• Congestive heart failure

- (ineffective pumping of blood through the circulatory system due to enlargement and weakening of the **heart** muscle) is the most common **cause** of **pleural effusion**. Pneumonia is a common lung infection and may result in **pleural effusion**.

- **Hypoalbuminemia**

is a medical sign in which the level of albumin in the blood is abnormally low.

Nephrotic syndrome

If there are too much protein losing from patient’s blood vessel, he will have more severe edema,

Hypoalbuminemia in NS can cause a decrease in oncotic pressure causing extravasation (leakage) of **fluid** into the interstitial space. In conditions of severe **hypoalbuminemia**, **fluid** extravasation may cause occurrence of **pleural effusion**.

Peritoneal dialysis

- complications in **PD** patients and result from the migration of **dialysis fluid** under pressure from the **peritoneal** cavity into the **pleural** space..

Types Of Effusions cont..

EXUDATIVE EFFUSIONS

A fluid rich in protein and cellular elements that oozes (leaking) out of blood vessels due to inflammation . It is caused by blocked blood vessels, inflammation, lung injury, and drug reactions. An exudate—which often is a cloudy fluid, containing cells and much protein .

Causes of Exudates:

- **Atelectesis –**

collapse of both lungs

Hemothorax Infection

- (bacteria, viruses, fungi, tuberculosis, or parasites)

Uremia

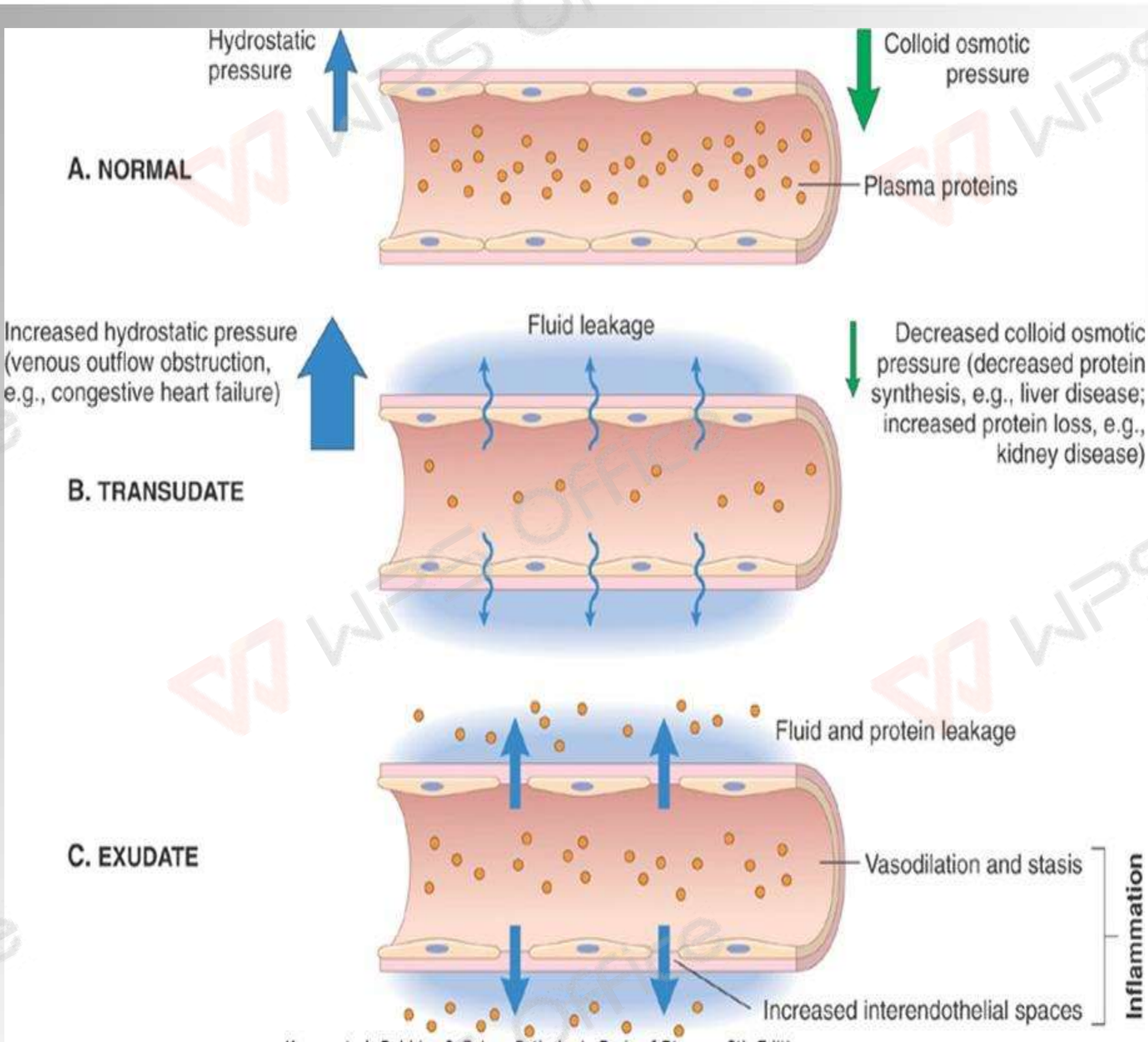
- fluid, electrolyte, and hormone imbalances and metabolic abnormalities,

Asbestos exposure

- These diseases can **lead** to irritation, swelling and inflammation, which in **causes** the blood vessels in the pleurae to leak extra **fluid** into the **pleural** space.

Pulmonary embolism

is a blockage in one of the **pulmonary** arteries in your lungs. In most cases, **pulmonary embolism** is caused by blood clots that travel to the lungs from the legs



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.

Pathophysiology

Due to etiological factors

It is explained by increased pleural fluid formation or decreased pleural fluid absorption

Increased pleural fluid formation can result from elevation of hydrostatic pressure (increasing weight of fluid) & decreased osmotic pressure.

It leads to increased capillary permeability (capacity of a blood vessel wall to allow for the flow of small molecules (drugs, nutrients, water, ions) or even whole cells & passage of fluid is through openings in the diaphragm

Hence production increases & absorption is decreases lymphatic obstruction Pleural effusions produce a restrictive ventilatory defect and also decrease the total lung capacity and vital capacity

Clinical Features

- **History :**
  - Dyspnea
  - Pleuritic chest pain
  - Cough
  - Fever
  - Hemoptysis
  - Wt. loss
  - Trauma
  - Hx. of cancer
  - Cardiac surgery
- **Physical :**
  - Dullness to percussion
  - Decreased breath sounds
  - Absent tactile fremitus
  - Other findings: ascites, JVP, peripheral edema, friction rub, unilateral leg swelling

CLINICAL MANIFESTATION

Pleuritic chest pain indicates inflammation of the parietal pleura Physical examination findings that can reveal the presence of an effusion dull or flat note on percussion

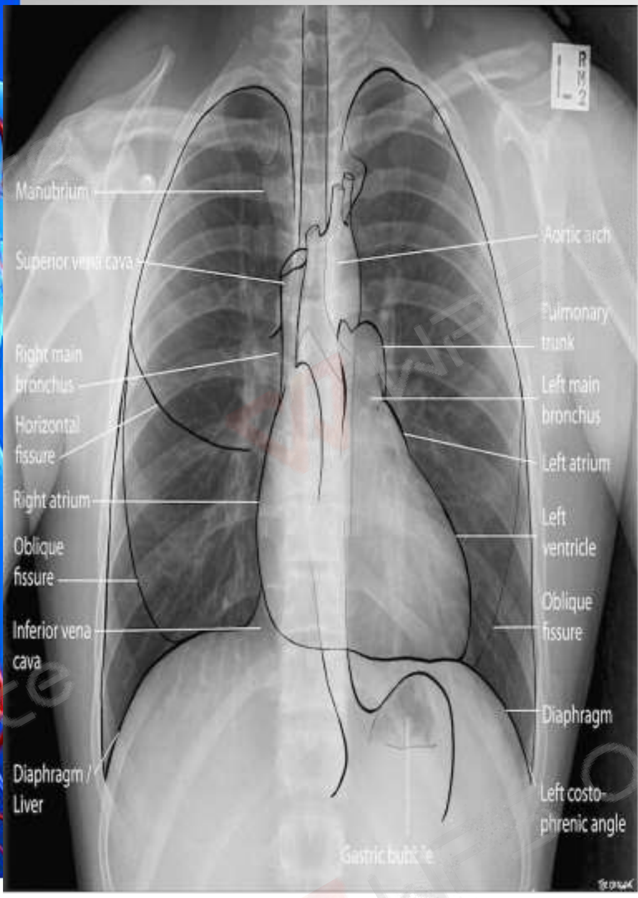
diminished or absent breath sounds on auscultation. Chest pain, usually a sharp pain that is worse with cough or deep breaths, Cough, Fever, Rapid breathing, Shortness of breath

DIAGNOSTIC EVALUATION

During a physical examination, the doctor will listen to the sound of your breathing with a stethoscope and may top on your chest to listen for dullness. The following tests may help to confirm a diagnosis : Chest CT scan Chest x-ray Pleural fluid analysis (examining the fluid under a microscope to look for bacteria, amount of protein, and presence of cancer cells) Thoracentesis (a sample of fluid is removed with a needle inserted between the ribs) Ultrasound of the chest

- **Chest Radiography :**The posteroanterior and lateral chest radiographs are still the most important initial tools in diagnosing a pleural effusion.
- **Ultrasound** is useful both as a diagnostic tool and as an aid in performing thoracentesis. It assist in identifying pleural fluid loculations.
- **Computed Tomography:** Cross-sectional computed tomography (CT) It helps distinguish anatomic compartments more clearly This modality is useful as well in distinguishing empyema

Normal CRG



Pleural effusion CRG

- MANAGEMENT

- Treatment depends on the cause of your pleural effusion and how bad your symptoms are. You may need any of the following:
- **Diuretics laxis** may help you lose extra fluid caused by heart failure or other problems.
- **Antibiotics** help prevent or treat an infection caused by bacteria.
- **Analgesic drug to relief pain**

- **NSAIDs** help decrease swelling and pain or fever..
- **Steroids** or other types of medicines may be given to decrease swelling.
- **Drainage** of extra pleural fluid may be done using thoracentesis or a chest tube. A chest tube may stay in your chest for days or weeks. This allows the extra fluid around your lungs to drain over time. You may need medicines put directly into your chest if the fluid does not drain out easily.

SURGICAL PROCEDURE

In some cases, the following may be done: Surgery

- **Thoracentasis** Pleural fluid is drawn out of the pleural space in a process called thoracentesis. A needle is inserted through the back of the chest wall in the sixth, seventh, or eighth intercostal space into the pleural space. The fluid may then be evaluated.
- **Gram stain and culture** to identify possible bacterial infections

Nursing Diagnosis
&
Nursing Intervention

- 1. Ineffective breathing pattern related to decreased lung expansion(accumulation of liquid), as evidenced by dyspnea, changes in depth of breathing, accessory muscle use.

Interventions

- Maintain a comfortable position is usually elevated headboard
- Given oxygen through a cannula (8mls)

2. Acute Pain related to accumulation of fluid in the pleural space and rubbing of thoracostomy tube to the lungs

- **Interventions**
- -The presence of pain, the scale and intensity of pain was well assessed
- -The client taught about pain management and relaxation with distraction
- -Chest tube secured to restrict movement and avoid irritation
- -Given prescribed analgesics i.e diclofenac 75mg.

3. Risk for nutrition impariment, less than body requirement related to inability to ingest adequate nutrients

Interventions

- -Patient relative i.e his father encouraged to give him energy reaching food stuff together with energy supplement so that he can get enough energy.
- -Administer DNS as prescribed to the patient to increase energy lost.

4. Risk for fluid volume deficit related to chest tube drainage.

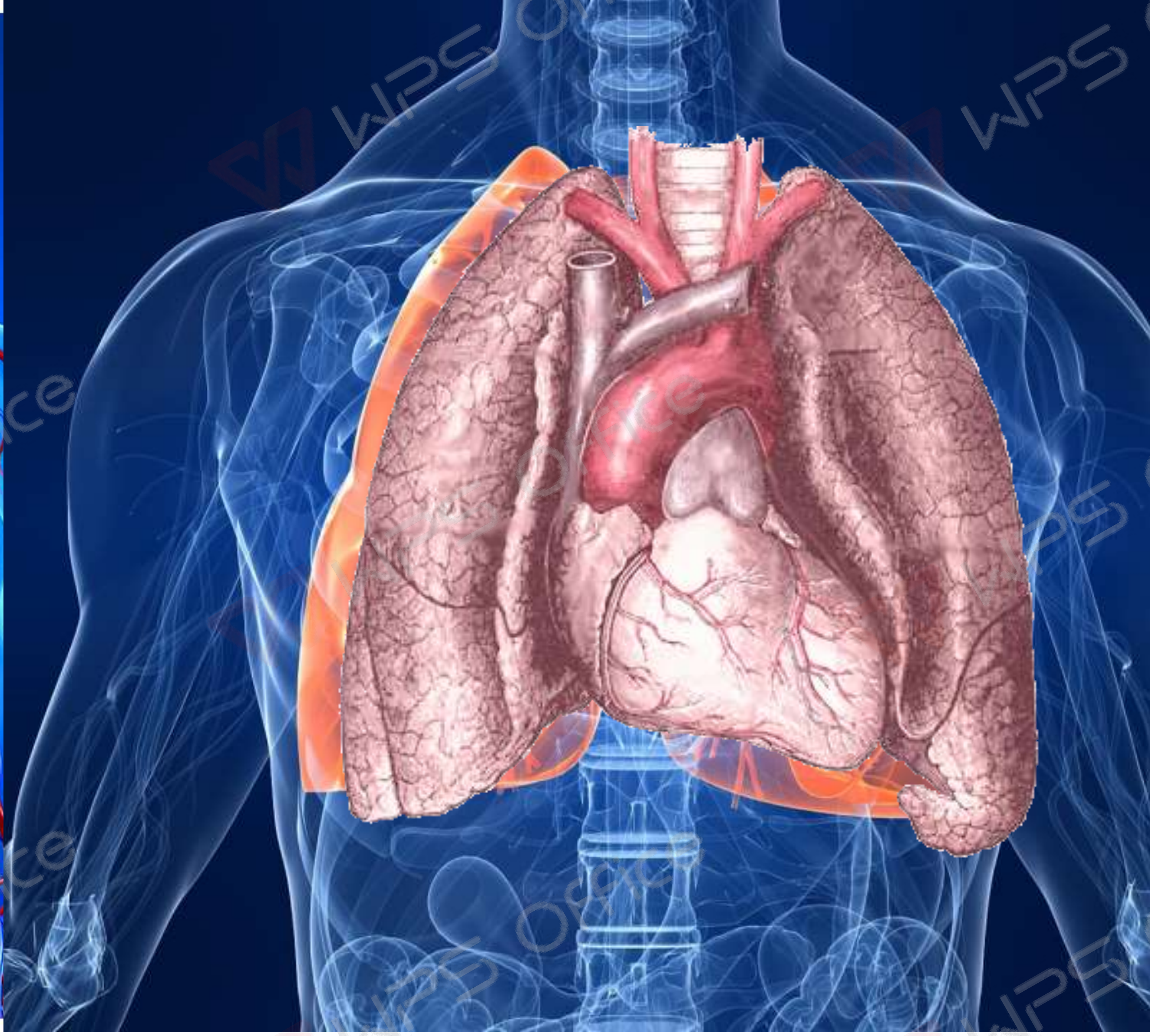
- **Interventions**
- -encourage the patient to drink enough water to supplement the one lost by chest tube drainage
- -IV fluids & DNS to replace fluid lost in drainage system monitored in 24hours.

5. Risk for infection related to the presence of fluid in the pleural space and the incision site.

- **Interventions**
- -The patient dressed at the incision site when it is wetted, probably after 2 to 3 days
- -Given antibiotics as prescribed i.e IV metronidazole 500mg 8 hourly, IV ceftiaxone 1gm.

Possible Complications

A lung that is surrounded by excess fluid for a long time may be damaged. Pleural fluid that becomes infected may turn into an abscess, called an **empyema**, which will need to be drained with a chest tube. Pneumothorax (air in the chest cavity) can be a complication of the thoracentesis procedure.


Thank
you!